A Negative Answer to the Double Arrow Ordinal Product Question

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Status. `full_solution_likely_valid`.

1 Source Question and Criterion

The source paper is Korpalski, *Semadeni-Pelczynski derivative and Banach spaces of continuous functions on nonmetrizable cubes*, arXiv:2502.16981 [1]. Let $\mathbb{S}$ denote the double arrow space. Question 1 on page 22 asks whether

$$C(\mathbb{S}) \simeq C(\mathbb{S} \times [0, \alpha])$$

for every ordinal $\omega^\omega \leq \alpha < \omega_1$.

Immediately after the question, Proposition 6.2 of the same source gives the following obstruction: if K is compact and $C(K)$ does not embed into $c_0(\mathfrak{c})$, then

$$C(\mathbb{S}) \not\simeq C(\mathbb{S} \times K).$$

> By Bessaga and Pełczyński's characterisation of isomorphisms of spaces of continuous functions on countable compacta [5], it follows that $C([0,\omega]) \not\simeq C([0,\alpha])$ for any ordinal $\omega^\omega \leq \alpha < \omega_1$. This raises the following question.
>
> **Question 1.** *Is the space $C(\mathbb{S})$ isomorphic to $C(\mathbb{S} \times [0,\alpha])$ for $\omega^\omega \leq \alpha < \omega_1$?*
>
> However, it is interesting to note that Proposition 6.1 does not hold if $[0,\omega]$ is replaced by any nonscattered compact space. Note that the proof of the following result is similar to that of Lemma 5.6.
>
> **Proposition 6.2.** *If K is a compact space such that $C(K)$ does not embed into $c_0(\mathfrak{c})$, then $C(\mathbb{S}) \not\simeq C(\mathbb{S} \times K)$.*
>
> *Proof.* Assume that $T : C(\mathbb{S}) \to C(\mathbb{S} \times K)$ is an isomorphism. There is a natural copy of $C([0,1])$ embedded in $C(\mathbb{S})$ (of functions constant on the second coordinate). Let $Y = T[C([0,1])]$; this is a separable subspace of $C(\mathbb{S} \times K)$. It is well known that the quotient $C(\mathbb{S})/C([0,1])$ is isomorphic to $c_0(\mathfrak{c})$; see [6, Section 3].
>
> Let us show that there exists an isomorphic copy of $C(K)$ inside $C(\mathbb{S} \times K)/Y$. Denote by D a countable dense subset of Y. For each $f \in C(\mathbb{S} \times K)$, define the section function $f_s(k) = f((s,k))$ for $s \in \mathbb{S}$, $k \in K$ and put

2 Key Observation

For $K = [0, \alpha]$ with $\omega^\omega \le \alpha < \omega_1$, the space $C(K)$ is separable but has Szlenk index strictly larger than that of c_0. Consequently $C(K)$ cannot embed into $c_0(\mathfrak{c})$.

The separability point is useful because every separable subspace of $c_0(\mathfrak{c})$ is contained in $c_0(A)$ for some countable $A \subseteq \mathfrak{c}$, hence in a copy of ordinary c_0. The Szlenk obstruction then applies in the separable space c_0.

3 Non-Embedding Lemma

Lemma 1. *Let $\omega^\omega \le \alpha < \omega_1$. Then $C([0, \alpha])$ does not embed isomorphically into $c_0(\mathfrak{c})$.*

Proof. Suppose that $J : C([0, \alpha]) \to c_0(\mathfrak{c})$ is an isomorphic embedding. The range $J[C([0, \alpha])]$ is separable. Choose a countable dense set D in this range. For $y \in c_0(\mathfrak{c})$, the support

$$\mathrm{supp}(y) = \{\gamma < \mathfrak{c} : y(\gamma) \neq 0\}$$

is countable. Hence

$$A = \bigcup_{y \in D} \mathrm{supp}(y)$$

is countable. Every coordinate outside A vanishes on D, hence vanishes on the closed range $J[C([0, \alpha])]$. Thus the range is contained in $c_0(A) \simeq c_0$.

By monotonicity of the Szlenk index under subspaces, this would give

$$\mathrm{Sz}(C([0, \alpha])) \le \mathrm{Sz}(c_0) = \omega.$$

But Brooker's computation of the Szlenk indices of ordinal intervals says that if γ is the unique ordinal satisfying

$$\omega^{\omega^\gamma} \le \alpha < \omega^{\omega^{\gamma+1}},$$

then

$$\mathrm{Sz}(C([0, \alpha])) = \omega^{\gamma+1}.$$

For $\alpha \ge \omega^\omega$ one has $\gamma \ge 1$, so this index is at least ω^2, a contradiction. □

4 Answer to Question 1

Theorem 1. *Let $\mathbb{S}$ be the double arrow space. For every ordinal $\omega^\omega \le \alpha < \omega_1$,*

$$C(\mathbb{S}) \not\simeq C(\mathbb{S} \times [0, \alpha]).$$

Thus Question 1 of [1] has a negative answer throughout the range in which it is asked.

Proof. Fix $\omega^\omega \le \alpha < \omega_1$ and put $K = [0, \alpha]$. By Lemma 1, $C(K)$ does not embed into $c_0(\mathfrak{c})$. Proposition 6.2 of the source paper therefore applies and gives

$$C(\mathbb{S}) \not\simeq C(\mathbb{S} \times K) = C(\mathbb{S} \times [0, \alpha]).$$

□

5 Verification Notes

The proof uses two external inputs. The first is Proposition 6.2 of the source paper. The second is the standard Szlenk-index computation for $C([0,\alpha])$ due to Brooker [2]. The reduction from $c_0(\mathfrak{c})$ to c_0 is included above: separable subspaces of $c_0(\mathfrak{c})$ have countable coordinate support.

The run indexes were searched for the arXiv id, the double-arrow keywords, the product question, $c_0(\mathfrak{c})$, Szlenk, and ordinal variants. Local source searches found no prior packet for this exact consequence. Web searches on 2026-06-29 found Brooker's Szlenk computation and no separate exact resolution of the double-arrow question.

6 Limitations and Review Recommendation

This is a counterexample-style resolution, not an independent reproving of Proposition 6.2. Human review should check that the target protocol permits using a later proposition from the source paper together with the classical Szlenk computation. Mathematically, the main points to verify are the countable-coordinate reduction inside $c_0(\mathfrak{c})$ and the cited Szlenk formula for $\alpha \geq \omega^\omega$.

References

[1] M. Korpalski, *Semadeni-Pelczynski derivative and Banach spaces of continuous functions on nonmetrizable cubes*, arXiv:2502.16981, 2025.

[2] P. A. H. Brooker, *Szlenk and w^*-dentability indices of ordinal $C(K)$ spaces*, arXiv:1210.3696, 2012.